REAL-ESTATE UNDERWRITING · PORTFOLIO CASE STUDY

DFW Rental Underwriting Model

Screen. Underwrite. Pass.

A screen-then-underwrite study of the Dallas–Fort Worth single-family rental market

Ajay Patil · Excel + Power Query · Data: Zillow ZHVI & ZORI (June 2026)

VERDICT: PASS AT 7%

The headline. 76110 Fort Worth, the strongest risk-adjusted pick of 166 ranked ZIPs, returns just **1.1% levered IRR** at a 7% mortgage, *below* its own 5.0% all-cash return. Cash flow is negative every year, DSCR never reaches 1.0, and rents would need to be about 23% higher to break even. The disciplined call is to **pass**: it clears an 8% levered hurdle only if the price falls about 15% (to ~$185K) at 7%, or about 10% (to ~$196K) with a 5–6% mortgage.

4.3%	1.1%	5.0%	1.10x	–6.1%	0.71
Entry cap rate	Levered IRR	Unlevered IRR	Equity multiple	Cash-on-cash, Yr 1	DSCR, Yr 1

01 Objective

I built an end-to-end model that underwrites a single-family rental in Dallas–Fort Worth and answers one question honestly: **at today's prices and a 7% mortgage, does buying a rental actually earn a return?** The project fills a specific gap in my portfolio: my Chipotle three-statement FP&A model and Python ML forecasting project don't compute investment-return math. This one does: cap rate, NOI, DSCR, cash-on-cash, levered and unlevered IRR, and equity multiple, over a full 10-year hold.

The design has two layers. First it **screens** the DFW ZIP universe on rental economics to find the most promising submarket; then it **underwrites** that ZIP in depth with a 10-year pro forma, a 360-month amortization schedule, two sensitivity tables, and a cap-rate exit. It is built in a single tool (Excel + Power Query) by design, with one firm constraint throughout: **no manufactured returns.** If the honest answer is "pass," that is the finding.

PROJECT SCOPE	
Market	**Dallas–Fort Worth–Arlington**
Asset	**Single-family rental**
Data	**Zillow ZHVI + ZORI (Jun 2026)**
Hold period	**10 years**
Selected ZIP	**76110 Fort Worth**

SKILLS DEMONSTRATED	
Core skill	**Real-estate underwriting**
Modeling	**10-yr pro forma + amortization**
ETL	**Power Query (M)**
Analysis	**Cap · DSCR · CoC · IRR · eq. mult.**
Stress testing	**Data Tables + Goal Seek**

02 Data & ETL

Power Query loads two national Zillow research files, home values (ZHVI) and rents (ZORI), straight from CSV, filters them to the Dallas–Fort Worth–Arlington metro, and unpivots each wide monthly series into a tidy long table (ZIP · City · County · Date · Value). ZIP codes are kept as text to preserve leading zeros. The data funnel is itself a finding: rent coverage is thinner than price coverage.

254	→	220	→	216	→	166
ZIPs with home values		ZIPs with rents		with both, screenable		full 3-yr history, ranked

03 Submarket screen, and a deliberate override

The 166 ZIPs with a full 3-year rent history are ranked on two independent signals, gross rental yield and 3-year rent CAGR (price-to-rent is the reciprocal of yield), combined into a composite weighted 70% yield / 30% growth. The top of the list looked like this:

#	ZIP	CITY	PRICE	RENT/MO	YIELD	3-YR CAGR
1	75216	Dallas	$177,032	$1,823	12.4%	4.5%
2	75215	Dallas	$177,515	$1,621	11.0%	3.4%
3	75134	Lancaster	$262,549	$2,251	10.3%	5.2%
4	**76110**	**Fort Worth**	**$217,756**	**$1,744**	**9.6%**	**6.2%**
5	75217	Dallas	$201,984	$1,769	10.5%	2.9%
6	75241	Dallas	$215,576	$1,710	9.5%	1.8%

Ranks #1–#3 were low-cost South Dallas and Lancaster ZIPs posting 10–12% "yields," the classic profile of distressed-area risk the data can't see (condition, taxes, tenant and vacancy risk). **I stepped past them and selected 76110 (rank #4):** a solid 9.6% yield paired with the shortlist's strongest rent growth, in a conventional, defensible submarket. Knowing when *not* to trust the model's top output is the analytical point.

04 Underwriting engine

A named-input assumptions block drives a 360-month amortization schedule (fully paying down to $0) and a 10-year pro forma. The exit is a terminal cap rate on forward NOI, net of selling costs and the loan payoff.

Gross rent → Vacancy → EGI → Tax · Insurance · Mgmt · Maint · CapEx → NOI → Debt service → Cash flow → Cap-rate exit

› Stress testing

Two live Excel Data Tables flex the result (exit cap × rent growth, and price × mortgage rate), and a Goal Seek solves for the rent that would zero out Year-1 cash flow, mapping exactly what would have to be true for the deal to work.

KEY ASSUMPTIONS	
Purchase price	**$217,756**
Down payment	**25% ($54,439)**
Mortgage	**7.0% · 30-yr fixed**
Rent (Yr 1)	**$1,744 / mo**
Rent growth	**3.0% / yr**
Vacancy	**5%**
Property tax	**2.24%, no homestead**
Insurance	**$2,000 / yr ***
Mgmt / Maint / CapEx	**8% / 5% / 5%**
Exit cap · Sell cost	**5.0% (+70bp vs entry) · 7%**

RESULTS (10-YR, LEVERED)	
Entry cap rate	**4.3%**
Levered IRR	**1.1%**
Unlevered IRR	**5.0%**
Equity multiple **	**1.10x**
Cash-on-cash (Yr 1)	**−6.1%**
DSCR (Yr 1 → 10)	**0.71 → 0.93**
Yr-1 cash flow	**−$3,718**
Cash flow, Yrs 1–10	**negative every yr**
Net sale proceeds	**$92,838**
Breakeven rent	**~$2,144 / mo**

* Insurance is a base-case placeholder below Fort Worth market averages (~$5,500/yr for $300K of dwelling coverage); each +$1,000/yr lowers levered IRR by ~5 points, which only strengthens the pass. ** Equity multiple counts the annual cash shortfalls as additional equity invested.

05 What the model found

1.1% vs 5.0%

Negative leverage

The asset yields ~4.3% unlevered, but the mortgage costs 7%. Financing a 4.3-cap asset with 7% debt *reduces* return: levered IRR lands well below unlevered.

0.71 → 0.93

DSCR below 1.0 for the entire hold

NOI never covers the mortgage. Lenders typically require 1.20+, so this wouldn't qualify for standard investment financing at these terms.

–$3,718

Negative carry, all ten years

Year-1 cash flow never turns positive (still –$877 in Year 10). The shortfall is funded out of pocket, and the return relies entirely on appreciation at exit.

~52% of NOI

Texas property-tax drag

At the full 2.24% non-homestead rate, Year-1 tax is $4,878: the largest expense (46% of OpEx) and the main reason a 9.6% gross yield shrinks to a 4.3% cap rate.

Conservative exit, same answer

The 5.0% exit cap is 70bp above the 4.3% entry cap. Exiting at the entry cap lifts levered IRR to only ~5.1%, still short of an 8% hurdle.

06 Levered IRR sensitivity: price × mortgage rate

Price ↓ · Rate →	5%	6%	7%	8%	9%
$185,093	13.3%	11.7%	10.0%	8.4%	6.7%
$195,980	10.6%	8.9%	7.2%	5.4%	3.6%
$206,868	7.9%	6.1%	4.2%	2.3%	0.3%
$217,756	5.1%	3.1%	1.1%	–1.1%	–3.3%
$228,644	2.2%	0.0%	–2.3%	–4.7%	–7.3%
$239,532	–0.9%	–3.5%	–6.1%	–9.0%	–12.0%

Base case: $217,756 at 7% · Clears 8% hurdle (my target for a leveraged SFR) · Positive, below hurdle · Negative

07 Conclusion: pass at 7%

76110 is the strongest conventional value the screen surfaced in DFW, and it still doesn't clear the bar. At today's price and rate it delivers roughly a 1% IRR, below risk-free Treasuries, in exchange for leverage, tenant and vacancy risk, illiquidity, and a decade of monthly out-of-pocket carry. Against an 8% levered hurdle, it becomes worth revisiting only if the price falls to **~$185K (–15%) at a 7% mortgage**, or to **~$196K (–10%) if rates ease to 5–6%**. The value of the exercise isn't a buy recommendation; it is a disciplined, evidence-backed *pass*.

08 What this project demonstrates

End-to-end build

Raw public data → automated ETL → quantitative screen → underwriting model → stress testing → a written recommendation, in one workbook.

Return math

Cap rate, NOI, DSCR, cash-on-cash, levered and unlevered IRR, and equity multiple over a 10-year hold.

Judgment over mechanics

Spotting a yield trap, using 3% forward rent growth instead of the recent 6.2% spike, and reaching an honest "pass."

Data: Zillow ZHVI & ZORI (rents, all rental types), ZIP level, June 2026 · Property tax checked against Tarrant Appraisal District rates · Educational portfolio project, not investment advice.